

B.E. Even Semester Midterm Examination, 2023-24(Even)

Subject: Design and Analysis of Algorithm | **Paper Code:** PCC-IT 404

Q1. Solve the recurrences (4+3+3)

i) $T(n) = 3T(n/4) + \Theta(n)$ -- Recursion Tree Root cost is cn . Level 1 cost is $3(c(n/4)) = (3/4)cn$. Sum over levels is a geometric series $cn \sum (3/4)^i$. Since ratio < 1 , it converges to a constant. Complexity is $\Theta(n)$. ii) $T(n) = 3T(n/4) + n \log n$ -- Master's Theorem $a=3, b=4$. $n^{\log_4 3} \approx n^{0.793}$. $f(n) = n \log n$ grows polynomially faster than $n^{0.793}$ (Case 3). Condition $3(n/4) \log(n/4) \leq k(n \log n)$ holds. Complexity: $\Theta(n \log n)$. iii) $T(n) = T(n-1) + 1$ -- Iteration Method $T(n) = T(n-2) + 2 = T(n-k) + k$. Base case $k=n \Rightarrow T(n) = T(0) + n$. Complexity: $\Theta(n)$.

Q2. Explain three basic asymptotic notations. (6)

Answer: 1. **Big O (O):** Upper Bound (Worst-case). $f(n) \leq c \cdot g(n)$. 2. **Big Omega (Ω):** Lower Bound (Best-case). $f(n) \geq c \cdot g(n)$. 3. **Theta (Θ):** Tight Bound. $c_1 \cdot g(n) \leq f(n) \leq c_2 \cdot g(n)$.

Q3. Worst-case time complexity of Quick Sort? Explain with recurrence. (4)

Answer: Worst-case is $O(n^2)$. Occurs when the array is already sorted and the pivot is the smallest/largest element, partitioning into sizes 0 and $n-1$. Recurrence: $T(n) = T(n-1) + T(0) + \Theta(n) \approx T(n-1) + cn$. Expanding yields $cn + c(n-1) + \dots = c \cdot n(n+1)/2 = \Theta(n^2)$.